

Kazakh British Technical University

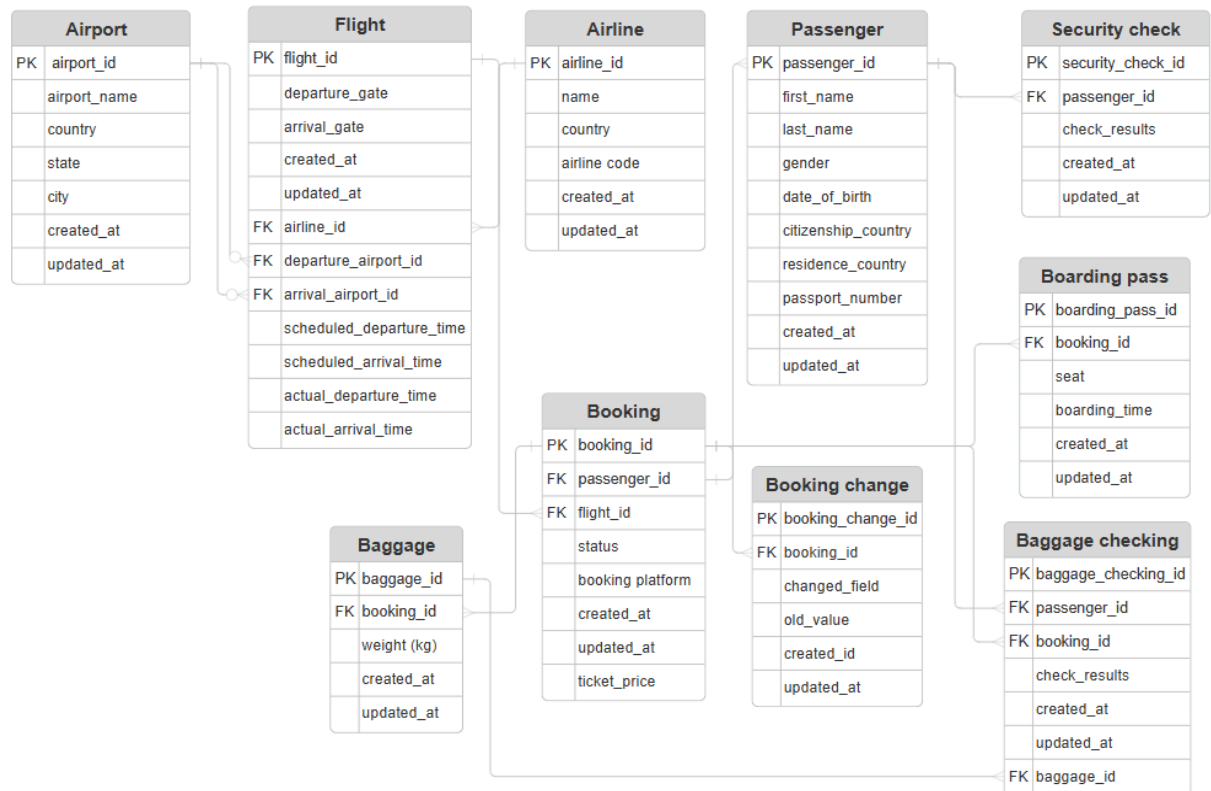
Data Bases

Laboratory Work №1

ERD Diagram

Full Name: Omar Aknaz Shadiarkyzy

16.09.2026
Almaty



Here is my ERD diagram. I made it at the site “SmartDraw”. In this diagram we have data base of the airport. It’s designed to manage different types of airport operations including passenger bookings, flights, airlines, and booking details.

Let’s start from airport. This entity stores information about airports, like identifier (Primary Key), airport name, country, state or region, city, creation and update dates. Because it is an independent entity, it has no foreign keys.

The Flight entity stores information about scheduled and actual flights: identifier (Primary Key), departure and arrival gates, scheduled and actual departure and arrival times, and creation and update dates. It contains three foreign keys: `airline_id` (link to Airline), `departure_airport_id` and `arrival_airport_id` (both linked to Airport, representing the departure and arrival airports). The Flight–Airline and Flight–Airport relationships are one-to-many (one airline or airport can have many flights).

The Airline entity stores information about airlines: identifier (primary key), airline code, name, country, as well as creation and update dates. It has no foreign keys, as it is an independent entity.

The Passenger entity stores information about passengers: identifier (primary key), first name, last name, gender, date of birth, country of citizenship, country of residence, passport number, and profile creation/update dates. It has no foreign keys, as it is an independent entity.

The Booking entity stores information about reservations made by passengers: identifier (primary key), status, booking platform, creation/update dates, and ticket price. It contains two foreign keys: `passenger_id` (link to Passenger) and `flight_id` (link to Flight). The Booking–Passenger and Booking–Flight relationships are one-to-many (one passenger or flight can have many bookings).

The Booking change entity stores the history of modifications made to a booking: identifier (primary key) the changed field, its old value, and creation or update dates. It contains one foreign key, `booking_id` (link to Booking). The Booking–Booking change relationship is one-to-many (one booking can have multiple recorded changes).

The Boarding pass entity stores information about boarding passes issued to passengers: identifier (primary key), seat, boarding time, and creation and update dates. It contains one foreign key, `booking_id` (link to Booking). The Booking–Boarding pass relationship is one-to-many (a booking may be associated with one or more boarding passes, for example if reissued).

The Baggage entity stores information about registered baggage: identifier (primary key), weight (kg), and creation/update dates. It contains one foreign key, `booking_id` (link to Booking). The Booking–Baggage relationship is one-to-many (one booking can include multiple pieces of baggage).

The Baggage checking entity stores the results of baggage inspections: identifier (primary key), check result, and creation/update dates. It contains two foreign keys: `booking_id` (link to Booking) and `passenger_id` (link to Passenger). Both the Booking–Baggage checking and Passenger–Baggage checking relationships are one-to-many.

The Security check entity stores the results of passenger security screenings: identifier (PK), check result, and creation/update dates. It contains one foreign key, `passenger_id` (link to Passenger). The Passenger–Security check relationship is one-to-many (one passenger can undergo multiple security checks).